# Mahutin Lucien VIDAGBANDJI

: LMAH, Université Le Havre Normandie, France.
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## Current Position

- **Temporary Teaching and Research Assistant (ATER)**, UFR Sciences et Techniques, Université Le Havre Normandie (Sept. 2025 – Août 2026).
  **Affiliation** : Laboratoire de Mathématiques Appliquées du Havre (LMAH).

## Research Interests

Statistical modelling, Extreme value theory, Quantile regression, statistical learning, simulation-based inference, Applied probability.

## Education

- **Ph.D. in Statistics** (March 20, 2026), LMAH, Université Le Havre Normandie, France.
  **Thesis title** : *Extreme Quantile Regression Based on Conditional GEV Models and Generalized Random Forests.*
  Supervisors : Alexandre Berred, Cyrille Bertelle, Laurent Amanton.
  Reviewers : Clément Dombry, Ghislaine Gayraud.
- **M.Sc. in Statistics and Probability** (2022), Institut de Mathématiques et des Sciences Physiques (IMSP), UAC.
- **B.Sc. in Fundamental Mathematics** (2019), Faculté des Sciences et techniques, UAC.

## Publications and Preprint

— Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. (2025). Generalized random forest for extreme quantile regression. *Communications in Statistics – Simulation and Computation*, 1–24. https://doi.org/10.1080/03610918.2025.2543854.

— Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. (2026). Penalized estimation of GEV parameters for extreme quantile regression. *Accepted in the Journal of Statistical Theory and Practice.* [Hal Preprint]

— Vidagbandji *et al.* (2026). Consistency of weighted maximum likelihood estimator for extreme quantile regression. *to be submitted.*

## Communications

1. Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. Local Weighted Maximum Likelihood Estimator for Extreme Quantile Regression. Atelier des doctorants des laboratoires LMI et LMRS, Université de Rouen, 10 Mars, 2026. (Invited)
2. Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. *Extreme quantile regression using generalized random forests and block maxima approach.* International Conference on Extreme Value Analysis, Probabilistic and Statistical Models and Their Applications (EVA), University of North Carolina at Chapel Hill, USA, June 22–27, 2025.

3. Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. *Penalized estimation of GEV parameters for extreme quantile regression.* 56^e Journées de la Société Française de Statistique, Campus Saint-Charles, Aix–Marseille Université, France, June 2–6, 2025.
4. Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. *Combining Extreme Value Theory and Random Forests for High Quantile Regression.* 17^e Journée de la Fédération Normandie Mathématiques, INSA Rouen Normandie, France, May 26, 2025.
5. Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. *Generalized random forest approach for GEV extreme quantile regression.* Rencontres des Jeunes Chercheurs Africains en France (6^th edition), Institut Henri Poincaré, Paris, France, December 12–13, 2024.
6. Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. *Generalized random forest for extreme quantile regression.* 26^th International Conference on Computational Statistics (COMPSTAT), University of Giessen, Germany, August 27–30, 2024.
7. Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. *Parameter estimation of the generalized extreme value distribution using generalized random forest methods.* 16^e Journée de la Fédération Normandie Mathématiques, Université de Rouen Normandie, France, July 5, 2024.
8. Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. *GEV-extremal random forest.* 55^e Journées de Statistique de la Société Française de Statistique (SFDS), Université de Bordeaux, Campus de la Victoire, France, May 27–31, 2024, pp. 1098–1105.
9. Vidagbandji, M. L., Berred, A., Bertelle, C., & Amanton, L. *Quantile regression : an approach based on the GEV distribution and machine learning.* French Regional Conference on Complex Systems, Université Le Havre Normandie, France, May 31–June 2, 2023, pp. 515–518.

## Academic Internships

- **May–August 2022** : Research internship (M2–IMSP).
  *Modeling spatial extremes using normal mean–variance mixtures.*
  Supervised by Aliou Diop.
- **June–August 2021** : Research internship (M1–IMSP).
  *M-estimators and Z-estimators.*
  Supervised by Carlos Ogouyandjou and Freedath DJIBRIL MOUSSA.

## Professional Experiences

- **2024–present** : Co-organizer of the LMAH PhD students' seminar, Université Le Havre Normandie, France.
- **December 2024** : Co-organizer of the MIIS and PSIME Doctoral School Day, IUT Le Havre, Quai Frissard, Université Le Havre Normandie, France.
- **September 2024** : Member of the organizing committee of the Bio Dynamics Days 2024 seminar, LMAH, Université Le Havre Normandie, France.
- **Since 2023** : Teaching Associate, UFR of Science and Technology, Université Le Havre Normandie, France.
- **May 2023** : Member of the local organizing committee of the French Regional Conference on Complex Systems (FRCCS 2023), LITIS, Université Le Havre Normandie, France.

## Teaching Experiences

- **2025–2026** :
  — Analysis 1 for 2^nd year undergraduate students in Mathematics and Computer Science (30 h).

— Probability for $2^{nd}$ year undergraduate students in Mathematics and Computer Science (20 h).
— Algebra for $1^{st}$ year undergraduate students in Mathematics and Computer Science (50 h).
— Basic Mathematical Tools 1 & 2 for $1^{st}$ year undergraduate students in Chemistry and Life Sciences ($2 \times 25$ h + 17 h).
— Mathematics for Computer Science for $2^{nd}$ year undergraduate students in Computer Science (30 h).

- **2024–2025** :
  — Practical sessions in LaTeX for $1^{st}$ year undergraduate students in Mathematics and Computer Science (6 h).
  — Basic Mathematical Tools 1 & 2 for $1^{st}$ year undergraduate students in Chemistry and Life Sciences (26 h).

- **2023–2024** :
  — Practical sessions in LaTeX for $1^{st}$ year undergraduate students in Mathematics and Computer Science (6 h).
  — Basic Mathematical Tools 1 & 2 for $1^{st}$ year undergraduate students in Chemistry and Life Sciences (26 h).

## Technical Skills

- **Programming :** R, Python, C.
- **Deep Learning :** PyTorch,

## Scholarships & Funding

- **2022–2025** : PhD project at LMAH, Université Le Havre Normandie, France.
- **2021–2022** : Master's scholarship recipient, African Center of Excellence (CEA–SMIA).
- **2017–2020** : Government scholarship of Benin for undergraduate studies, UFR Sciences et Techniques, Université d'Abomey-Calavi, Benin.